Wednesday November 18th

9:00-9:30 **Welcome to IVCNZ 2026**

9:30-10:30 **Keynote: Dr Kevin Shedlock**

Seeing With, Not Seeing For: Computer Vision and Indigenous Knowledge

10:30-11:00 **Morning Tea**

11:00-12:15 **Oral Session: Detection & Recognition**

Weng, Yu-Chian; Lin, Hung-I; Yeh, Hao-Yun; Sun, Min-Te; Sakai, Kazuya; Ku, Wei-Shinn *Small Object Detection based on Spatial-Frequency Extraction and Attention-Guided Gated Fusion*
Auer, Franziska; Kuruppanthara Mukundan, Parvathi; Meisen, Tobias *Lightweight Attention Enhancements for YOLOv7 Sonar Object Detection*
Hines, Corey; Green, Richard *Real-Time Detection of Parking Lot Occupancy*

12:15-13:30 **Lunch**

NZ Robotics and Automation Society AGM

13:30-15:15 **Oral Session: Best Paper Nominees**

Kaur, Asavri; Singh, Sonit; Hassanzadeh, Tahereh; Ho Shon, Ivan; Sowmya, Arcot *Context-aware Global-attention based Deep Learning model for Lymphoma Tumour Segmentation in FDG-PET/CT scans*
Templeton, Charlie; Mills, Steven; O'Regan, Gerard *Deep Learning for Māori Rock Art: Detection, Classification, and Generalisation*
Ciceklidag, Pasa; Mian, Ajmal; Hassan, Mubashar; Hong, Jin *RoadTwin: A Training-Free Drivable Hybrid Digital Twin of a 114 km Freeway from Mobile LiDAR*
Rogers, Mitchell; Fischer, Johannes; Debski, Igor; Mischler, Claudia; Delmas, Patrice; Xue, Bing; Zhang, Mengjie *Counting and Verifying Nesting Albatross in Satellite Images*

15:15-15:45 **Afternoon Tea**

15:45-17:00 **Online Session**

Tan, Shengsheng; Zhong, Liezhou; Lewis, Joshua R.; Shah, Syed Afaq Ali *Hierarchy-Risk-Aware Reference Gallery Curation for Fine-Grained Produce Recognition*
Rehmat, Sadaf; Iqbal, Jamshaid; Jiyun, Li; Iman, Tuba; Noor, Fatima *Aerotrack-net: A Unified Architecture for Joint Detection and Tracking of Drones vs. Birds*
Li, Hang; Fan, Jianan; Cai, Weidong *Semi-VGM: A Semi-Supervised Pixel-Level Contrastive Learning Framework for Medical Image Segmentation*
Hassanzadeh, Tahereh; Singh, Sonit; Yang, Junyi *Topology- and Geometry-Aware 3D Circle of Willis Segmentation from CTA and MRA*
Hassanzadeh, Tahereh; Singh, sonit; wu, Zizhang; Sowmya, Arcot *Multimodal CTA–MRA Fusion for 3D Circle of Willis Segmentation Across Reference Spaces*
Younas, Umair; Ur Rehman, Saeed; Anas, Qazi *CRIS-Net: Continuous Resolution-agnostic Implicit Segmentation for Camouflaged Objects*

17:30-19:00 **Conference Reception, St Margaret's College**

Thursday November 19th

9:00-9:30 **Online Session**

Patel, Shrey *Defect-Aware Masked Autoencoder Approach to Unsupervised Steel Surface Defect Recognition*
Mäenpää, Timo; Bappy, Akash; Ariram, Siva; Bavirisetti, Durga Prasad; Tikanmäki, Antti *RangeGen PoleSight: A Benchmark Dataset and Scalable Range-Image Generation Framework for Roadside Pole Detection*
Ghosh Chowdhury, Arnab; Smith, Amos; Atzmueller, Martin *Analyzing the Influence of MobileViT Blocks on Robustness to Image Corruption and Texture Reduction in Land Use Classification*

9:30-10:30 **Keynote: Prof Tanja Alderliesten**

Beyond Accuracy: What It Takes for Medical Imaging AI to Succeed in Clinical Practice

10:30-11:00 **Morning Tea**

11:00-12:30 **Oral Session: Applications**

Lu, Yu; Deng, Xiaokang; Zhang, Wei; Huang, Bingding *EchoST-SAM: Semantic-Prompted Spatiotemporal LV Segmentation for Functional Analysis in*
Mozhaeva, Anastasia; Fedorov, Vladimir; Chisholm, Kyle A.; Vinetsky, Vsevolod; Delmas, Patrice *Motion- and Perception-Adaptive Video Compression for Real-Time UAV Piloting*
Fan, Xiang; Rogers, Mitchell; Zhao, Junhong ; Xue, Bing; Zhang, Mengjie *Cross-Zoo Transfer of Chimpanzee Pose Estimation and Chimp–Object Interaction Recognition*
Macdonald, Isaac; Green, Richard *Automated Rubik's Cube State Extraction and Solving Using Deep Learning and Adaptive Colour Classification*

12:30-14:00 **Lunch**

IVCNZ Steering Group Meeting

14:00-14:30 **Poster 'Barking' Session**

Kumar, Pratul; Jain, Nandini; Sharma , Shaurya; Tyagi, Ashish; Kumar, Ajay; Kumar Dalela, Dr. Pankaj *QuAVF: Quality-Aware Adaptive Face Verification Under Degradation*
Wang, Haoming ; Chakraborti, Tapabrata ; Choudhury, Zakia Zinat; McCane, Brendan; Coffey, Sean *Deconvolutional Vision Mamba in Crossmodal Medical Image Synthesis*
Liang, Sendong; Nand, Parma; Yan, Wei Qi *DeepSymbol: A Multimodal Framework for Cultural Symbol Understanding*
Lee, Nick; Schofield, Sam; Bainbridge-Smith, Andrew *ORB-SLAM3: Preventing Catastrophic Failure on Resource-Constrained Configurations*
Liu, Chuning; Yan, Wei Qi *Parameter-Efficient Transfer Learning for Image Fragment Reassembly: Robustness Analysis of CNN and Vision Transformer Approaches*
Guo, Xuan; Zhao, Junhong ; Xue, Bing; Zhang, Mengjie; Casanovas, Paula; Graham, Shaun; Hopkins, Grant *Learning Discriminative Features via Spatially Guided Attention for Animal Image Classification with Small Data*
Mao, Weixi; McCane, Brendan ; Deng, Jeremiah *Foundation-Model-Assisted Semi-Automatic Annotation for Fish Tracking in Ecological Videos*

Peng, Hao; March de Ribot, Francesc; Zhao, Junhong *Few-Shot CHRPE Classification: A Retinal Image Dataset and Comparative Evaluation*
Rogers, Steven; Weti, Tainui; Delmas, Patrice ; Mozhaeva, Anastasia *PSNR-VR: A Full-Reference Quality Metric for Wide-Angle VR Video Incorporating Human Visual System Models*
Hwang, jaeyeon; Swain Ranjan, Bishal; Ko, Jaepil *Agentic reconstruction of mathematical diagram from scanned images*
Baron, Dominic; Green, Richard *Rowing Technique Analysis using Pose Detection*
Lindsay, Oscar; Green, Richard *Real-Time Face Swap*
Storey, Adam; Green, Richard *Gait Analysis for Dairy Cow Lameness Detection*
Huang, Yu-Chuan; Green, Richard *Accelerating Fabric Defect Detection in Videos while Improving Accuracy*

14:30-15:45 **Posters and Afternoon Tea**

15:45-17:00 **Online Session**

Hafeez, Huma; Garratt, Matthew; Plested, Jo; Iyer, Sankaran; Sowmya, Arcot *Count-Constrained Association for Real-Time Multi-Object Tracking in Fixed-Camera Panoramic Video*
Nandakumar, Adithya; George, Stellin John; Agarwal, Aditya; James, Arvin; J, Angel Arul Jothi; Rajasekar, *MalVisionNet: A Lightweight Image-Based Malware Detection Framework using CNNs for IoT Devices*
Adn, Taqiyudin; Bachtiar, Fitra *Validation-First Evaluation of Mobile Backbones and Paired Knowledge Distillation for Efficient Crowd Counting*
Islam, Rafiul; Sumayya, Rihila; Ahmed, Adib; Kushol, Rafsanjany; Kabir, Md. Hasanul *Neuro-Duo-MIL: A Dual-Branch DINOv3 Framework for Leakage-Aware Alzheimer's Disease Classification Using 2D MRI and Clinical Data*
Zia Ur Rehman, Muhammad; Aamir, Zuha; Janjua, Naeem ; Ul Islam, Shams *Cross-Scene Generalisation for Joint HSI-LiDAR Classification via Frozen Multimodal Representations*
Datta, Rohit; Islam, Syed *A Critical Review of Deep Learning Methods for Cardiovascular Risk Stratification from Retinal Fundus Images*

18:00-21:00 **Conference Dinner, Tūhura Otago Museum**

Friday November 20th

9:00-9:30 **Online Session**

Jin, Qiyuan; Li, Huilin; Barodawala, Fakhruddin; Findlay, Rebecca; Black, Joanna; Turuwhenua, Jason *CGTA-CNN: A Lightweight Channel-Gated Temporal-Attention CNN for Participant-Independent Optokinetic Nystagmus Detection*
Li, Huilin; Jin, Qiyuan ; Barodawala, Fakhruddin ; Black , Joanna ; Findlay, Rebecca ; Turuwhenua, Jason *High-accuracy Few-shot Model for Pupil Localization*

9:30-10:30 **Keynote: Dr Philip McLeod**

From Pixels to Poles: GeoCam's Vision-Based Pipeline for Mapping Infrastructure

10:30-11:00 **Morning Tea**

11:00-12:15 **Oral Session: Art, Culture, and Heritage**

Templeton, Charlie; Mills, Steven *Baseline Networks for 3D Lithic Fragment Part Segmentation*
Zhan, Haolan; Xu, Zhiwei; Frermann, Lea; Drummond, Tom; Lau, Jey Han *Syn-Det: A Large-Scale Benchmark for Generalizable AI-Generated Image Detection*
Hayes, Samantha; Green, Richard *Analysis of Turns in Artistic Roller Skating*

12:15-13:15 **Lunch**

(NZ Robotics and Automation Society AGM)

13:15-14:30 **Oral Session: Vegetation**

Lin, Yida; zhang, mengjie; xue, bing; green, richard; schofield, sam *Choosing a Pseudo-Ground-Truth Generator for Tree-Branch Depth Estimation: A Zero-Shot Benchmark of Deep Stereo Matching for Forestry UAVs*
Lloyd Jones, Laurie; Whaanga, Hēmi; Varaine, Kerian; Mills, Steven *Reconstructing and Rendering Virtual Vegetation: Comparing Plant Models in Virtual Reality*
Scott, Bradley ; Schofield, Sam; Green, Richard *Reachability-Driven Cut Selection and Ordering Towards UAV-Based Tree Pruning*

14:30-14:45 **Conference Closing**

15:00-16:30 **HCI and Visual Computing Lab Tour**